PARISHRAM 2027

Electric Charges and Fields

LECTURE-2 PHYSICS

Rakshak sir



Topics To Be Covered

1. Charges - properties ✓
2. Methods of charging ✓
3. Coulomb's Law ...
4. ...


Rakshak Sir
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@RAKSHAKSIRSCIENCETG



2026-27 EDITION

CBSE 2027 EXAMINATION

QUESTION & CONCEPT BANK

CHAPTER-WISE & TOPIC-WISE



- Chapter-wise CONCEPT MAPS
- Important Questions with Detailed Explanations NCERT & EXEMPLAR
- Handpicked & High yield from Past 10 Years PYQs
- Revision Blue Print & Solved Questions COMPETENCY FOCUSED
- CBSE 2026 Past Year Solved Papers LATEST CBSE PAPERS
- MOCK TESTS As per Latest Pattern

QCB

CLASS 12

PHYSICS

Rakshak Dua

Class XII (2025-2026)

Physics (Theory)

Time: 3 hrs. **Max Marks: 70**

		Marks
Unit-I	**Electrostatics**	
	Chapter-1: Electric Charge and fields	5
	Chapter-2: Electrostatic Potential and Capacitance	5
Unit-II	**Current Electricity**	
	Chapter-3: Current Electricity	6

Electric Charges and Fields

Electric charges. Conservation of charge, Coulomb's law-force between two-point charges, forces between multiple charges; superposition principle and continuous charge distribution. Electric field, electric field due to a point charge, electric field lines, electric dipole, electric field due to a dipole, torque on a dipole in uniform electric field.

Electric flux, statement of Gauss's theorem and its applications to find field due to infinitely long straight wire, uniformly charged infinite plane sheet and uniformly charged thin spherical shell (field inside and outside).

Electric Charge

→ 'q' or 'Q'

Its is an intrinsic property of the elementary particles like electrons, protons, etc. of which all the objects are made up of.

Its is a scalar quantity.

It SI unit is Coulomb (C).

e^-

p^+

n^0 → neutral
uncharged ✗

$1\,mC = 10^{-3}\,C$

$1\,\mu C = 10^{-6}\,C$

$1\,nC = 10^{-9}\,C$

$1\,\beta C = 10^{-12}\,C$

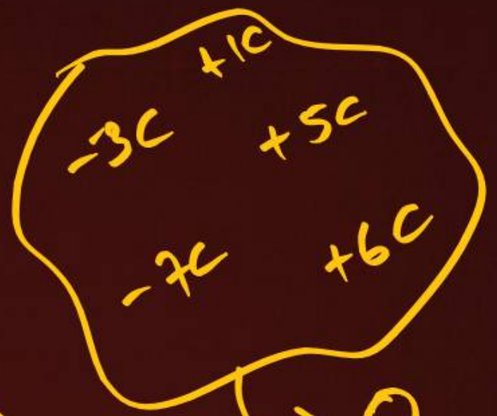
'e' = 1.6×10^{-19} coulomb

charge on an electron / proton

electronic charge

Basic Properties of Charge

1. Additivity ⇒ Scalar property
2. Quantization ⇒ $1e^- = 1.6 \times 10^{-19} C$, $2e^- = 3.2 \times 10^{-19} C$, $3e^- = 4.8 \times 11$, $4e^- = 6.4 \times 11$, $5e^- = 8 \times 10^{-19}$
3. Conservation ⇒ const.
4. Charge is invariant
5. Accelerated Charge radiates Energy



$Q_{net} = 6+5+1-3-7$
$= +2C$

$2 \times 10^{-19} C$ → NOT POSSIBLE

Class 9th
Y.K.B.

ne
Natural No.



Basic Properties of Charge

Addition of Charges

If a system contains three point charges q_1, q_2 and q_3, then the total charge of the system will be algebraic addition of q_1, q_2 and q_3, i.e., charge will add up

$$q = q_1 + q_2 + q_3$$

Basic Properties of Charge

2) Quantization of charges

Electric charge is always quantized, i.e., electric charge is always an integral multiple of charge 'e'.

$q = ne$, where $n = 0, \pm 1, \pm 2, \ldots\ldots\ldots$

q = total charge

n = no. of electrons

e = electronic charge

$1e$
$2e$
$3e$
$4e$
$5e$
$6e$
$\vdots$
ne

Basic Properties of Charge

3) Conservation of charges

Law of conservation of charge

1. The total charge of an isolated system remains constant.
2. The electric charges can neither be created nor destroyed, they can only be transferred from one body to another. (More details later)

Basic Properties of Charge

4. Charge is invariant

Charge is independent of frame of reference. i.e., charge on a body does not change whatever be its speed.

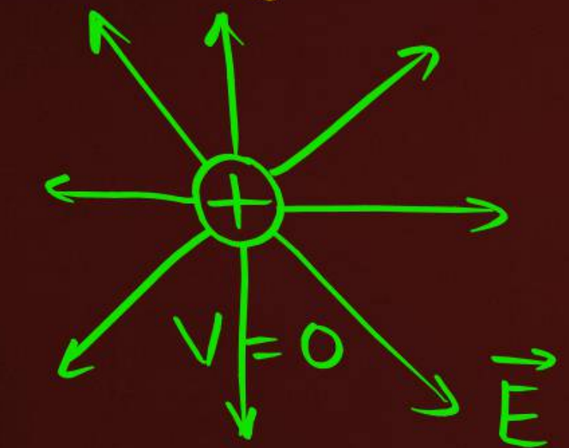
length
mass
time
} Change with speed Comparable to speed of light.

Charge } Const.

Basic Properties of Charge

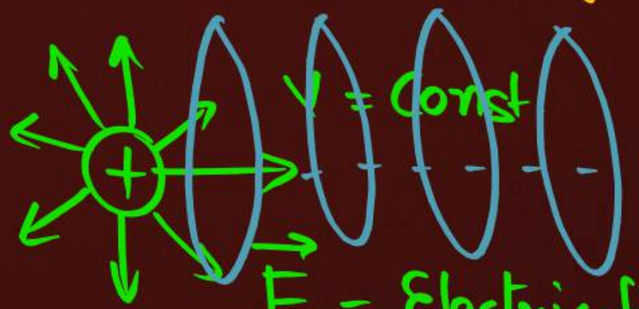
5. Accelerated Charge Radiates Energy

1) Charge at Rest



Electric field ✓

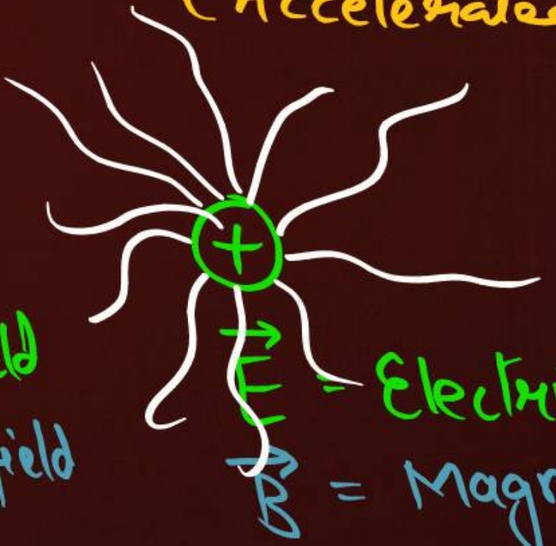
2) Charge at Uniform Motion (Constant velocity)



$\vec{E}$ = Electric field

$\vec{B}$ = Magnetic field

3) Charge in non-uniform (Accelerated)

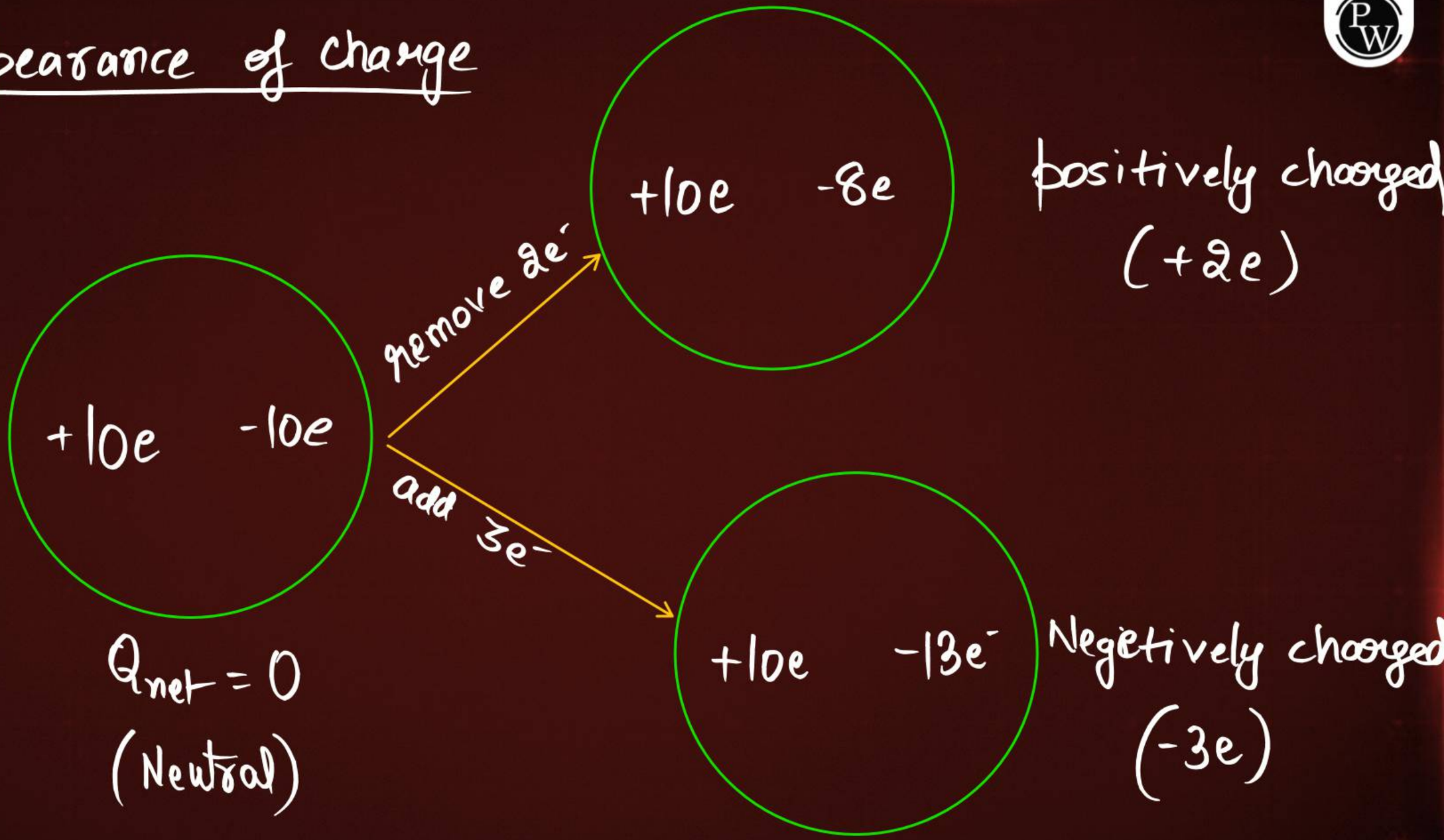


$\vec{E}$ = Electric

$\vec{B}$ = Magnetic

EM Waves

* Appearance of Charge



Methods of Charging

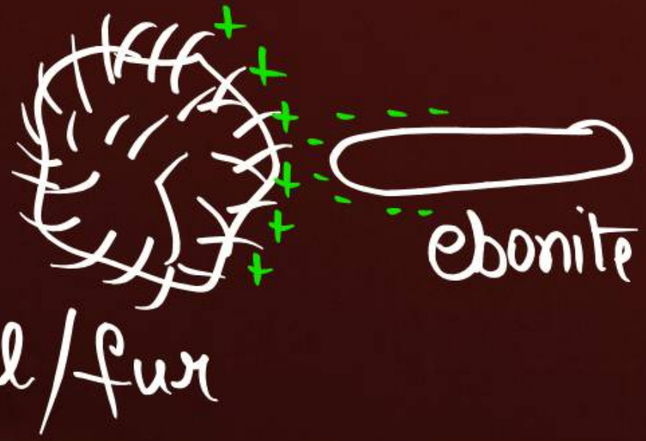
① By rubbing / friction :-

i) Silk cloth & Glass Rod



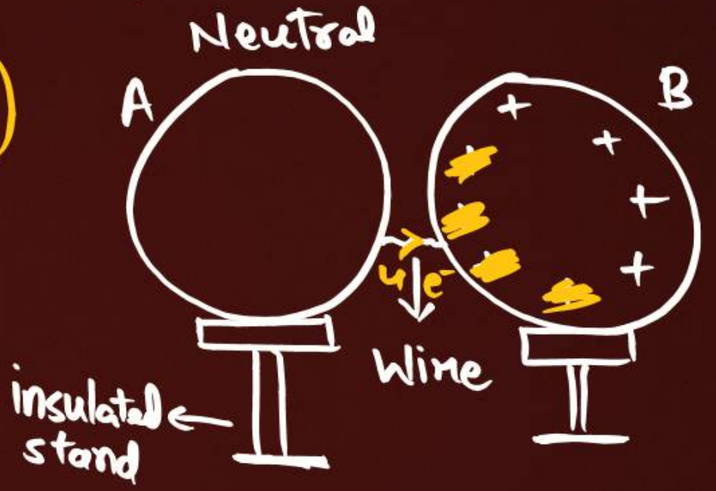
ii) Woolen cloth/fur & ebonite rod

RDx
e^- = ebonite
↳ Negative



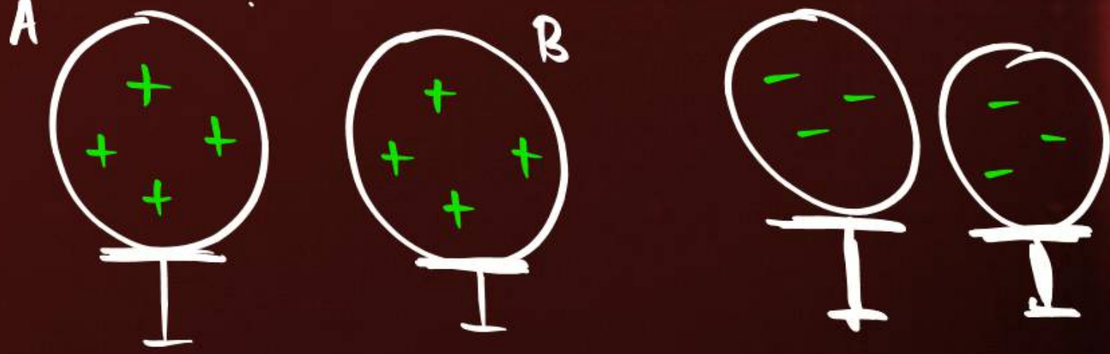
② By Conduction (for Conductors) (Physical touch)

i)



ii)





HOMEWORK

- DPP
- Recall karke soye

Thank You